Milestone 2 – Segmentation Performance Report

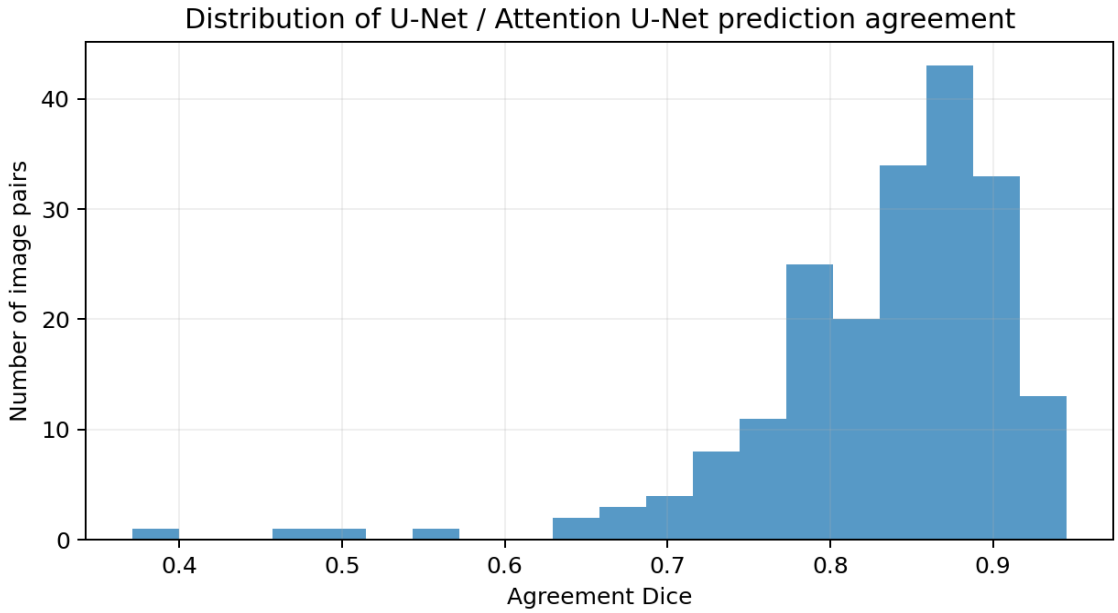
Project: Artery Segmentation and Stenosis Detection

Basis: CSV and JSON results supplied in artery_model_code.zip. Agreement between predictions is distinguished from accuracy against ground truth.

1. Prediction agreement

Paired masks: 200. Mean agreement Dice: 0.8309. Mean agreement IoU: 0.7182.

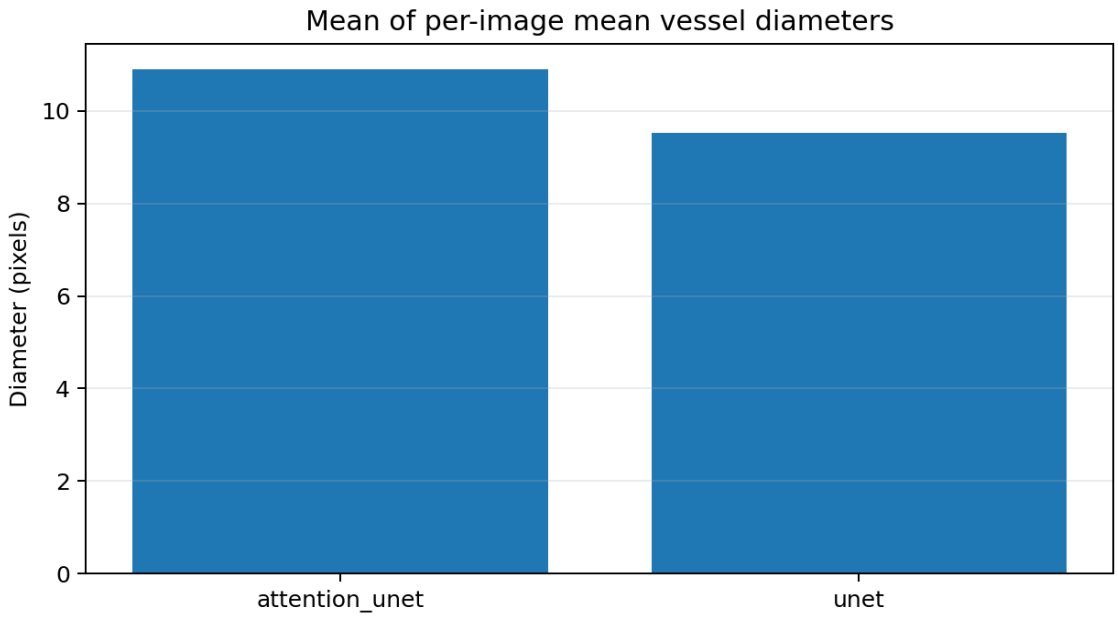
These metrics measure overlap between U-Net and Attention U-Net predictions, not accuracy against expert annotations.



2. Vessel diameter summary

Statistics are calculated across image-level records, giving each image equal weight.

Measure	U-Net	Attention U-Net	Unit
Images	200	200	count
Total measured centerline points	162047	148104	points
Mean of per-image mean diameter	9.52	10.90	px
Median of per-image median diameter	8.79	10.20	px
Mean of per-image P90 diameter	14.81	16.30	px



3. Limitations

• No ground-truth masks were used in the supplied prediction-agreement calculation.

• Ground-truth Dice, IoU, precision, recall, and specificity cannot be established from these files alone.

• Diameter estimates are in pixels; millimetre conversion requires image calibration.

• Prediction agreement does not establish which model is more accurate.

4. Conclusion

The supplied files support initial comparison of prediction agreement and vessel-diameter characteristics for 200 images per model. Ground-truth masks and verified image-to-mask correspondence are still required for a true segmentation accuracy report.